The higher processing capabilities of devices and newer algorithms have brought about a radical change in the list of possibilities in the technology world. Catching up with this trend are programming related hardware. Combining hardware programming with software programming will enable making unimaginable tasks possible.

Why do we need more advanced technology?

Let's look at a real world example.

Say we need to design a system that assists the blind to move around. This is possible using the current version of Arduino™ and an ultrasonic sensor, which detects how far objects are from you. By sensing this distance and informing a blind person whether he's near or far away from a wall, this system can make it possible for a blind person to move around.

Time to replace the stick of the blind

But there are limitations to this approach.

- The proper function of this system depends on the ultrasonic sensor, which is itself a very unreliable device.
- This sensor has a range sensing capability of 30-100 centimetres and could go up to 150-200 centimetres. Anything lower than 30 cms and beyond 100 cms gives wrong readings, so special attention must be given to detect such cases.
- Since an ultrasonic sensor can only detect objects directly in front, you'll need to move around to detect objects on either side.
- The mentioned hardware can work in places like a home and office, which have no moving objects. Bigger problems start to arise when a blind person wishes to navigate along roads.
- When crossing a road, there are chances of an ultrasonic sensor detecting no vehicle. Due to the sensor's limited range, a fast moving vehicle even a bit beyond its range might go undetected. This could lead to serious consequences. There's no way a blind person can benefit from a road signal.